

Revised Load Interconnection System Impact Study Report

For Queue Project Q19-1168

Morris Cogeneration Llc 230 kV
October 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1168. The Interconnection Customer proposes a 635.1 MW large load (data center) facility interconnecting at the Morris Cogeneration Llc 230 kV substation in the Illinois area. The request was received on March 12, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 357 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,659,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Morris Cogeneration Llc-Dresden Generating Station 230kV	Morris Cogeneration Llc-Boca Raton 230kV	123.4	622	45.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Morris Cogeneration Llc-Dresden Generating Sta 230kV	Rebuild line with higher-capacity conductor	\$7,968,000	33% (\$2,659,000)
	Total allocated to project		\$2,659,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.